MIL-C-915/50A
1 August 1972
SUPERSEDING 1/
MIL-C-915/50(SHIPS)
20 March 1972

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 300 VOLTS, TYPE MS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS

Qualification required.

Construction

First - Copper conductor, AWG No. 16 (7/.0192) or ASTM B-286 Size 16-19, tin coated. Maximum diameter 0.063 inch.

Second - Polyvinyl chloride insulation, 105°C. white. Minimum thickness 0.008 inch.

Third - Clear polyamide jacket on each conductor, type III, grade E, or type IV of MIL-M-20693, nominal thickness 0.003 inch. Standard Identification Code, applied by Method 1.

Fourth - The required number of singles (see details) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with No. 1, from the center outward.

Fifth - Binder tape, polyester, type G of MIL-I-631, applied helically with overlap.

Sixth - Shielding braid of AWG No. 34 or No. 36 tin coated copper. Minimum coverage of 85 percent.

Seventh - Standard thermoplastic jacket (gray). Cable jacket thickness 0.060 inch min. Cable surface marking required.

Details

Type and size	No. of condrs.	Overall diameter	
		Min. (inch)	Max. (inch)
MS-37	37	0.740	0.800

NOTE: Weight of completed cable shall not exceed 0.60 pounds per foot.

EXAMINATION AND TESTS

	Requirements:
Basic electrical:	
Conductor resistance - Ohms/1000 ft. at 25°C., maximum	5.1
Voltage withstand - Volts, rms, minimum	
Conductor to conductor	2000
Conductor to shield	1000
Insulation resistance - Megohms/1000 ft. minimum	500
Group A:	
Visual and dimensional	

1/ Requirements formerly covered on MIL-C-24145/10.

A Denotes changes

FSC 6145

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	Requirements:
Group B:	
(A) Cold bending, cable - At minus 30° ± 2°C. mandrel diameter 10 inches, maximum	No damage
Physicals (unaged)	
Insulation (conductor)	
Tensile strength - psi, minimum	1800
Elongation - Percent, minimum	100
Jacket (cable)	
Tensile strength - psi, minimum	1800
Elongation - Percent, minimum	225
Group C:	
Physicals (aged)	
Insulation (conductors)	
Air oven	
Elongation - Percent of unaged, minimum	90
Jacket (cable)	
Hot oil immersion	
Tensile strength - psi, minimum	1400
Elongation - Percent, minimum	50
Permanence of printing (conductor) - Cycles, minimum	50
Permanence of printing (jacket) - Cycles, minimum	250

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B and C.

UNIT ORDERING LENGTHS: 1000 feet (nominal)

(A) Custodians:
Army - MI
Navy - SH
Air Force - 80

(A) Review activities:
Army - EL, MI, AV
Navy - EC
Air Force - 80

(A) User activities:
Army - ME, MU, WC
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-0609-40)

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